

## Question 16 – Stage 1

Two convergent geometric sequences,  $\{u_1, u_2, u_3, \dots\}$  and  $\{v_1, v_2, v_3, \dots\}$ , of positive terms (common ratio,  $0 < r < 1$ ) are determined by the following values:

$$u_1 = 6, u_3 = 4 \text{ and } v_1 = 7, v_2 = 5.$$

The difference,  $D$ , between their sums to infinity is  $\left| \frac{6}{1-\sqrt{\frac{2}{3}}} - \frac{7}{1-\frac{5}{7}} \right| = 8.197$  (to 4 sig.figs)

Place the integers 2 to 7 in the boxes below, using each integer once, to determine two convergent geometric sequences of positive terms for which the value of  $D$  is as large as possible.

$$u_1 = \square, u_{\square} = \square \qquad v_1 = \square, v_{\square} = \square$$

What is this maximum value of  $D$  to 4 sig.figs?

### Working

$$\begin{aligned} 7r_v &= 5 \\ r_v &= \frac{5}{7} \end{aligned}$$

$$\begin{aligned} 6r_u^2 &= 4 \\ r_u^2 &= \frac{2}{3} \\ r_u &= \frac{\sqrt{6}}{3} \end{aligned}$$

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In [3]: from itertools import permutations
from pyTools import roundSF

def findR(u1,un,unValue):
    return (unValue / u1)**(1/(un-1))

def sumToInfinityGeometric(a,r):
    return a/(1-r)

possibleD = []

for arrangement in permutations(range(2,8)):
    u1, un, unValue, v1,vn,vnValue = arrangement

    # If it is converging
    if unValue < u1 and vnValue < v1:
        ur = findR(u1,un,unValue)
        vr = findR(v1,vn,vnValue)

        uSum = sumToInfinityGeometric(u1,ur)
        vSum = sumToInfinityGeometric(v1,vr)

        possibleD.append(abs(uSum-vSum))

possibleD.sort()

print(f"Maximum D: {roundSF(possibleD[-1],4)}")

Maximum D: 186.8
```